1 $n - 1$, n and $n + 1$ are any three consecutive integers.

(i) Show that the sum of these integers is always divisible by 3.

[1]

(ii) Find the sum of the squares of these three consecutive integers and explain how this shows that the sum of the squares of any three consecutive integers is never divisible by 3.

[3]

2 You are given that n, $n + 1$ and $n + 2$ are three consecutive integers.

(i) Expand and simplify $n^2 + (n + 1)^2 + (n + 2)^2$.

[2]

(ii) For what values of n will the sum of the squares of these three consecutive integers be an even number? Give a reason for your answer.

[2]

3

(i) Disprove the following statement:

$$3^n + 2 \text{ is prime for all integers } n \geqslant 0.$$

[2]

(ii) Prove that no number of the form 3^n (where n is a positive integer) has 5 as its final digit.

[2]

4

(i) Factorise fully $n^3 - n$.

[2]

(ii) Hence prove that, if n is an integer, $n^3 - n$ is divisible by 6.

[2]

5 Either prove or disprove each of the following statements.

(i) 'If m and n are consecutive odd numbers, then at least one of m and n is a prime number.'

[2]

(ii) ' If m and n are consecutive even numbers, then mn is divisible by 8.'

[2]

6 You are given that n is a positive integer.

By expressing $x^{2n} - 1$ as a product of two factors, prove that $2^{2n} - 1$ is divisible by 3.

[4]

7

Suppose x is an irrational number, and y is a rational number, so that $y = \frac{m}{n}$, where m and n are integers and $n \neq 0$. Prove by contradiction that $x + y$ is not rational.

[4]

8 Given that arcsin x = arccos y, prove that $x^2 + y^2 = 1$. [Hint: Let arcsin $x = \theta$] **[3]**

9 By finding a counter example, disprove the following statement.

If p and q are non-zero real numbers with $p < q$, then $\frac{1}{p} > \frac{1}{q}$. **[2]**

10 The spreadsheet in Fig. 5 shows a multiplication table. The numbers 35, 36, 35 in the shaded cells are of the form $n^2 - 1$, n^2, $n^2 - 1$, where $n = 6$. This pattern can also be seen for the other square numbers on the diagonal of the table in cells B2, C3, …, I9.

	A	B	C	D	E	F	G	H	I	J
1	1	2	3	4	5	6	7	8	9	10
2	2	4	6	8	10	12	14	16	18	20
3	3	6	9	12	15	18	21	24	27	30
4	4	8	12	16	20	24	28	32	36	40
5	5	10	15	20	25	30	35	40	45	50
6	6	12	18	24	30	36	42	48	54	60
7	7	14	21	28	35	42	49	56	63	70
8	8	16	24	32	40	48	56	64	72	80
9	9	18	27	36	45	54	63	72	81	90
10	10	20	30	40	50	60	70	80	90	100

Fig. 5

The spreadsheet can be extended to include larger numbers. Prove that the pattern holds for all integers $n > 1$. **[3]**

11 OACB is a parallelogram. O is the origin and point A has coordinates (5, 6). Point B has position vector $\mathbf{b} = -2\mathbf{i} - 7\mathbf{j}$.

(a) Find the coordinates of point C. [3]

M is the midpoint of AB.

(b) Prove that $\overrightarrow{OM} = \overrightarrow{MC}$. [3]

(c) Find the exact distance MC. [2]

12 (a) Prove the identity $\sec\theta - \cos\theta \equiv \tan\theta \sin\theta$. [4]

(b) Hence or otherwise solve the equation $\sec\theta - \cos\theta = \frac{1}{2}\tan\theta$ for $0 \leq \theta < 2\pi$. [4]

13 Prove from first principles that the derivative of x^3 is $3x^2$. **[5]**

14 (a) **In this question you must show detailed reasoning.**

Determine the exact values of k for which the curves $y = x^2 - kx$ and $y = 3(k + 1) + kx - x^2$ touch. **[6]**

(b) Determine whether or not there is a value of k for which the curves cross on the y-axis. **[4]**

15 Use a counter example to disprove the following statement.

$$2^n - 1 \text{ is prime for all } n > 1$$

[2]

16 The smallest of three consecutive positive integers is n. Find the difference between the squares of the smallest and largest of these three integers, and hence prove that this difference is four times the middle one of these three integers. **[4]**

17 You are given that the sum of the interior angles of a polygon with n sides is $180(n - 2)°$. Using this result, or otherwise, prove that the interior angle of a regular polygon cannot be 155°. **[3]**

18 Fig. 4 shows a cuboid *ABCDEFGH*. You are given that $FA > BC > AB > 0$. The distance between *F* and *C* is 5 cm. Given that the lengths *AB* and *BC* are both integers, prove that the length *FA* cannot be an integer. **[3]**

Fig. 4

19 Fig. 7 shows an isosceles triangle ABC, in which AC = BC = a and angle ACB = θ.

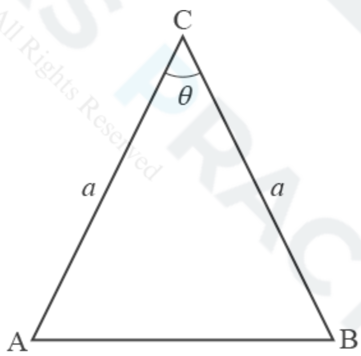


Fig. 7

(a) Use the cosine rule to write down an expression for AB^2 in terms of a and θ. **[1]**

(b) D is the midpoint of AB. Use an expression for AD in terms of *a* and θ to show that

$$\cos\theta \equiv 1 - 2\sin^2(\tfrac{1}{2}\theta).$$

[3]

The identity in part **(b)** is valid for all values of θ. However, the argument in parts **(a)** and **(b)** does not prove the identity for all values of θ.

(c) State the values of θ for which the argument is valid. **[1]**

The equation of a curve is $y=\frac{a}{(x+b)^2}$. Fig. 5 shows the curve for particular values of the constants *a* and *b*.

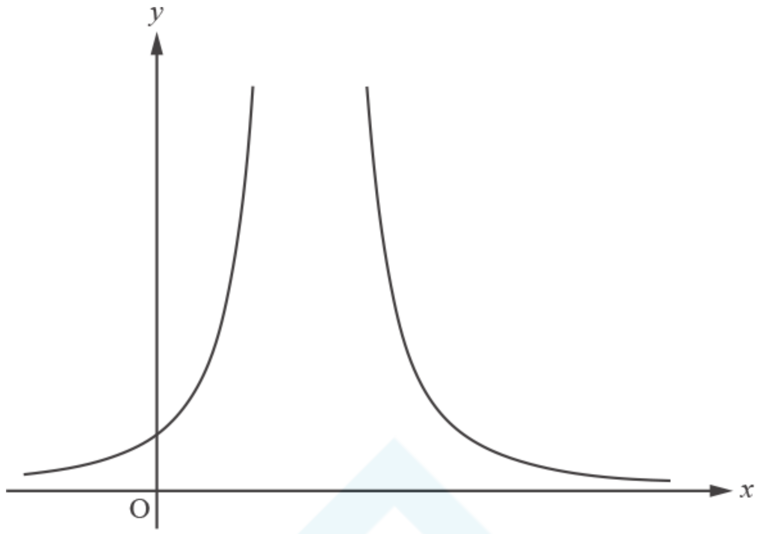


Fig. 5

(a) Write down the equations of the asymptotes, in terms of *a* and/or *b* where necessary. **[2]**

(b) Joe says "For all values of *a* and *b*, the curve lies above the *x*-axis." Determine whether Joe's statement is true or false. **[2]**

(c) The curve goes through the points (1, 3) and (3, 3). Find the values of a and b. **[4]**

21 P and Q are consecutive **odd** positive integers such that $P > Q$.

Prove that $P^2 - Q^2$ is a multiple of 8. **[3]**

22 Rose and Emma each wear a device that records the number of steps they take in a day. All the results for a 7-day period are given in Fig. 7.

Day	1	2	3	4	5	6	7
Rose	10 014	11 262	10 149	9361	9708	9921	10 369
Emma	9204	9913	8741	10 015	10 261	7391	10 856

Fig. 7

The 7-day mean is the mean number of steps taken in the last 7 days. The 7-day mean for Rose is 10 112.

(a) Calculate the 7-day mean for Emma. **[1]**

At the end of day 8 a new 7-day mean is calculated by including the number of steps taken on day 8 and omitting the number of steps taken on day 1. On day 8 Rose takes 10 259 steps.

(b) Determine the number of steps Emma must take on day 8 so that her 7-day mean at the end of day 8 is the same as for Rose. **[4]**

In fact, over a long period of time, the mean of the number of steps per day that Emma takes is 10 341 and the standard deviation is 948.

(c) Determine whether the number of steps Emma needs to take on day 8 so that her 7-day mean is the same as that for Rose in part **(b)** is unusually high. **[3]**

23 Fig. 12 shows the circle $(x-1)^2 + (y+1)^2 = 25$, the line $4y = 3x - 32$ and the tangent to the circle at the point A (5, 2). D is the point of intersection of the line $4y = 3x - 32$ and the tangent at A.

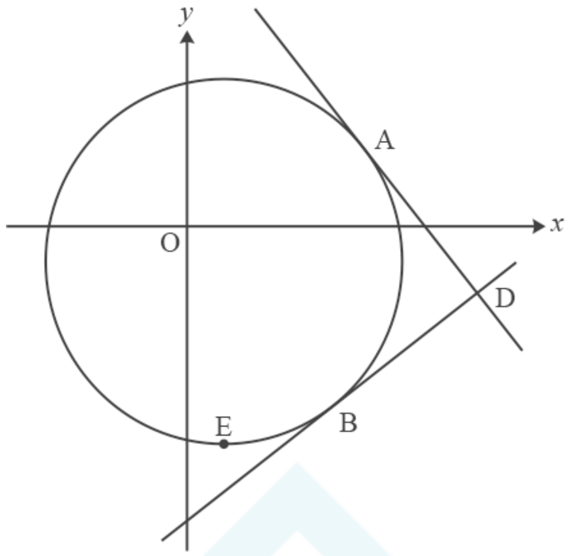


Fig. 12

(a) Write down the coordinates of C, the centre of the circle. [1]

(b) (i) Show that the line $4y = 3x - 32$ is a tangent to the circle. [4]

(ii) Find the coordinates of B, the point where the line $4y = 3x - 32$ touches the circle. **[1]**

(c) Prove that ADBC is a square. **[3]**

(d) The point E is the lowest point on the circle. Find the area of the sector ECB. **[5]**

(See Insert for Jun18 64003.) Lines 5 and 6 outline the stages in a proof that $\frac{a+b}{2} \geqslant \sqrt{ab}$. Starting from $(a - b)^2 \geq 0$, give a detailed proof of the inequality of arithmetic and geometric means. **[3]**

25 (See Insert for Jun18 64003.)

(a) In Fig. C1.3, angle CBD = θ. Show that angle CDA is also θ, as given in line 23. **[2]**

(b) Prove that $h = \sqrt{ab}$, as given in line 24. **[2]**

26 (See Insert for Jun18 64003.) It is given in lines 31 – 32 that the square has the smallest perimeter of all rectangles with the same area. Using this fact, prove by contradiction that among rectangles of a given perimeter, $4L$, the square with side L has the largest area. **[3]**

27 Prove that $x^2 + x + 2 > 1$ for all real values of x. **[3]**

Starting from the formula Price elasticity of demand $= \dfrac{\text{\% increase in quantity demanded}}{\text{\% increase in price}}$, as given on line 19, show that, at point A in Fig. C1 the price elasticity of demand is $\dfrac{P}{mQ}$, where m is the gradient of the straight line. **[3]**

29(a) **In this question you must show detailed reasoning.**

Nigel is asked to determine whether $(x + 7)$ is a factor of $x^3 - 37x + 84$. He substitutes $x = 7$ and calculates $7^3 - 37 \times 7 + 84$. This comes to 168, so Nigel concludes that $(x + 7)$ is not a factor.

Nigel's conclusion is wrong.

- Explain why Nigel's argument is not valid.
- Show that $(x + 7)$ is a factor of $x^3 - 37x + 84$. **[2]**

(b) Sketch the graph of $y = x^3 - 37x + 84$, indicating the coordinates of the points at which the curve crosses the coordinate axes. **[5]**

(c)

The graph in the above part is translated by $\begin{pmatrix}1\\0\end{pmatrix}$. Find the equation of the translated graph, giving your answer in the form $y = x^3 + \alpha x^2 + bx + c$ *where* α, b *and* c *are integers.* **[4]**

30 **[3]**

Without using a calculator, prove that $3\sqrt{2} > 2\sqrt{3}$.

31 A student's attempt to pr ove by contradiction that there is no largest prime number is shown below.

If there is a largest prime, list all the primes. Multiply all the primes and add 1. The new number is not divisible by any of the primes in the list and so it must be a new prime.

The proof is incorrect and incomplete.

Write a correct version of the proof. **[3]**

32 Celia states that $n^2 + 2n + 10$ is always odd when n is a prime number.

Prove that Celia's statement is false. **[2]**

33(a) **Please refer to Insert H640-03 Maths B November 2020 to answer this question.**

Show that if a = I and b > I then $a^b > b^a$. **[2]**

(b) Find integer values of a and b with b > a > I and a^b not greater than b^a (a counter example to the conjecture given in lines 7–8). **[1]**

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Part marks and guidance	
1		i	$3n$ isw	1	accept equivalent general explanation	
		ii	at least one of $(n-1)^2$ and $(n+1)^2$ correctly expanded	M1	must be seen	M0 for just $n^2 + 1 + n^2 + n^2 + 1$
		ii	$3n^2 + 2$	B1		accept even if no expansions / wrong expansions seen
		ii	comment e.g. $3n^2$ is always a multiple of 3 so remainder after dividing by 3 is always 2	B1	dep on previous B1 B0 for just saying that 2 is not divisible by 3 – must comment on $3n^2$ term as well allow B1 for $\frac{3n^2+2}{3} = n^2 + \frac{2}{3}$ **Examiner's Comments** A small number of candidates made no attempt to generalise in either part, and simply gave examples to demonstrate the properties, so, of course, gained no marks. Most earned the mark in the first part for adding the values. In the second part there were some errors in multiplying but many correctly reached $3n^2$ +2. The last mark was often lost due to an incomplete explanation centred on the fact that 2 was not divisible by 3, without making any reference to the fact that $3n^2$ is always divisible by 3.	SC: n, $n + 1$, $n + 2$ used similarly can obtain first M1, and allow final B1 for similar comment on $3n^2 + 6n + 5$
			Total	4		
2		i	$3n^2 + 6n + 5$ isw	B2	M1 for a correct expansion of at least one of $(n + 1)^2$ and $(n + 2)^2$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	odd numbers with valid explanation	B2	marks dep on 9(i) correct or starting again for B2 must see at least odd × odd = odd [for $3n^2$] (or when n is odd, $[3]n^2$ is odd) and odd [+ even] + odd = even soi condone lack of odd × even = even for $6n$; condone no consideration of n being even or B2 for deductive argument such as: $6n$ is always even [and 5 is odd] so $3n^2$ must be odd so n is odd B1 for odd numbers with a correct partial explanation or a partially correct explanation or B1 for an otherwise fully correct argument for odd numbers but with conclusion positive odd numbers or conclusion negative odd numbers B0 for just a few trials and conclusion **Examiner's Comments** The straightforward algebra in the first part was done correctly by most candidates. The most common errors were to write $(n + 1)^2$ as $n^2 + 1$, or sometimes $n^2 + 2n + 2$ and $(n + 2)^2$ as $n^2 + 4$. There was some encouraging work in the proof part with a number of	accept a full valid argument using odd and even from starting again ignore numerical trials or examples in this part – only a generalised argument can gain credit

Question			Answer/Indicative content	Marks	Part marks and guidance	
					slightly different methods being demonstrated. The majority considered the three terms that they had found in (i) but others went further and expressed the quadratic function as $3n(n + 2) + 5$ or $3(n + 1)^2 + 2$. As these were the more capable candidates they were then often argued the case elegantly. Some candidates returned to the original function successfully with a few replacing n by $2m + 1$ and expanding to find a factor of 2. The candidates who fared badly were those who failed to draw any conclusion at all, those who attempted to use an incorrect expression from (i) or those who just tried to show it with some numerical values.	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
3		i	$3^5 + 2 = 245$ [which is not prime]	M1	Attempt to find counter-example	If A0, allow M1 for $3^n + 2$ correctly
		i		A1	correct counter-example identified **Examiner's Comments** This proved to be very straightforward, with nearly everyone quoting the first correct counter-example of 245 (though a few came up with some much larger numbers).	evaluated for 3 values of n
		ii	$(3^0 = 1)$, $3^1 = 3$, $3^2 = 9$, $3^3 = 27$, $3^4 = 81$, …	M1	Evaluate 3^n for $n = 0$ to 4 or 1 to 5	allow just final digit written
		ii	so units digits cycle through 1, 3, 9, 7, 1, 3,			
		ii	…	A1		
		ii	so cannot be a ‘5’. OR			
		ii	$3n$ is not divisible by 5	B1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	all numbers ending in ‘5’ are divisible by 5. so its last digit cannot be a ‘5’	B1	must state conclusion for B2 **Examiner's Comments** This was not quite so easy as part (i). Most candidates who got full marks spotted the cyclic pattern in the units digits of 3^n as n increases. However, a significant minority evaluated 3^n for n = 0 to 9 and then cited ‘proof by exhaustion’. The second approach, less commonly used, was to use the fact that numbers ending in ‘5’ must be multiples of 5, and 3^n contains no factors of 5. However, many candidates who used this approach were unable to express the argument clearly enough and made incorrect statements.	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
4		i	$n^3 - n = n(n^2 - 1)$	B1	two correct factors	
		i	$= n(n - 1)(n + 1)$	B1	**Examiner's Comments** Many candidates failed to factorise the $n^2 - 1$, leaving their answer as $n(n^2 - 1)$. This rendered the second part of the question very difficult.	
		ii	$n - 1$, n and $n + 1$ are consecutive integers	B1		
		ii	so at least one is even, and one is div by 3 [$\Rightarrow n^3 - n$ is div by 6]	B1	**Examiner's Comments** There were two ideas needed here, the realisation that $n - 1$, n and $n + 1$ were consecutive integers, and that the product contained factors 2 and 3. Many candidates argued that the product had to be even, but this was not enough to gain credit. Others, predictably, verified the result with a few values of n, often describing this as 'proof by exhaustion'.	
			Total	4		

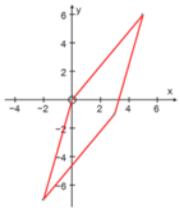
Question			Answer/Indicative content	Marks	Part marks and guidance	
5		i	False e.g. neither 25 and 27 are prime	B1	correct counter–example identified	Need not explicitly say 'false'
		i	as 25 is div by 5 and 27 by 3	B1	justified correctly **Examiner's Comments** Most candidates stated this was false and looked for a counter-example, usually 25 and 27. We did require them to show their counter-examples were composite for a second B mark.	
		ii	True: one has factor of 2, the other 4, so product must have factor of 8.	B2	or algebraic proofs: e.g. $2n(2n + 2) = 4n(n + 1) = 4 \times$ even × odd no so div by 8 **Examiner's Comments** This was less successful. Most candidates could see it was true, but then failed to come up with a coherent argument. Some wrote $2n(2n+2)= 4n^2 + 4n$ or equivalent, but then failed to explain why this is then divisible by 8 (rather than just 4, which got 1 out of 2). Most successful candidates got the idea that alternate even numbers are divisible by 4 and hence the product of this with another even number is divisible by 8.	B1 for stating with justification div by 4 e.g. both even, or from $4(n^2 + n)$ or $4pq$
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
6			$x^{2n} - 1 = (x^n - 1)(x^n + 1)$	B1		
			one of $2^n - 1$, $2^n + 1$ is divisible by three $2^n - 1$, 2^n and $2^n + 1$ are consecutive integers;	M1		award notwithstanding false reasoning condone 'factor' for 'multiple'
			one must therefore be divisible by 3;	A1		
			but 2^n is not, so one of the other two is	A1	if justified, correct reason must be given	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			2^n is not div by 3, and so has remainder 1 or 2 when divided by 3; if remainder is 1, $2^n - 1$ is div by 3; if remainder is 2, then $2^n + 1$ is div by 3 [so $2^{2n} - 1$ is divisible by 3]	A2		Examiner's Comments The first B1 for factorising $x^{2n} - 1$ was well done, but convincing proofs of the divisibility of $2^{2n} - 1$ by 3 were few and far between. We awarded M1 if candidates recognised that either $2^n - 1$ or $2^n + 1$ were divisible by 3, and two 'A' marks for proving this. The next 'A' mark was gained for stating that the consecutive numbers $2^n - 1$, 2^n and 2^{n+1} must include a multiple of 3, and the final mark for stating that 2^n is **not** divisible by 3; however, many candidates wrongly stated that 2^n was even and therefore not divisible by 3, or that two consecutive odd numbers must include a multiple of 3. The most elegant alternative solution seen was: $x^{2n} - 1 = (x^2 - 1)(x^{2n-2} + x^{2n-4} + \ldots + 1) \Rightarrow 2^{2n} - 1 = (2^2 - 1)(2^{2n-2} + 2^{2n-4} + \ldots + 1) = 3m$, where m is an integer. The language used by candidates in their explanations was often rather imprecise. In particular, the terms 'factor' and 'multiple' were often used incorrectly.
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
7			Suppose $x + y$ is rational So $x+y=\frac{p}{q}$, where p and q are integers $\Rightarrow\ x=\frac{p}{q}-\frac{m}{n}=\frac{(pn-mq)}{qn}$ which is rational x is irrational so this is a contradiction	E1(AO2.1) B1(AO2.1) B1(AO3.1a) E1(AO2.4) [4]	or stating that the difference of two fractions is rational	
			Total	4		
8			$\arcsin x = \theta$ $\Rightarrow x = \sin\theta$ $\arccos y = \theta \Rightarrow y = \cos\theta$ $\sin^2\theta + \cos^2\theta = 1$ $\Rightarrow x^2 + y^2 = 1$ AG	M1(AO1.1) M1(AO1.1) E1(AO2.1) [3]		
			Total	3		
9			E.g. $p = -1$, $q = 2$ $\frac{1}{p}=-1$, $\frac{1}{q}=\frac{1}{2}$ So $\frac{1}{p}<\frac{1}{q}$ for these values.	B1(AO3.1a) E1(AO2.1) [2]	correct counter example stated shown	
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance		
10			Use of $n + 1$ and $n - 1$ $(n + 1)(n - 1) = n^2 - 1$ Therefore (the conjecture is) true for all positive integers n (greater than 1)	B1(AO 2.1) M1(AO 1.1a) E1(AO 2.1) [3]	For both expressions seen Must be multiplied with attempt to expand brackets Clear conclusion must be stated		
			Total	3			

Question			Answer/Indicative content	Marks	Part marks and guidance	
11		a	 a = 5i + 6j $\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC} = \overrightarrow{OB} + \overrightarrow{OA} = \mathbf{b} + \mathbf{a}$ $= (5\mathbf{i} + 6\mathbf{j}) + (-2\mathbf{i} - 7\mathbf{j}) = 3\mathbf{i} - \mathbf{j}$ So C is (3, –1)	M1(AO 1.1a) M1(AO 3.1a) E1(AO 3.2a) [3]	Using vector notation Using **a** for vector $\overrightarrow{BC}$ in an addition Must be coordinates; do not allow for position vector only	
		b	$\overrightarrow{OM} = \frac{1}{2}(\mathbf{a} + \mathbf{b}) = \frac{3}{2}\mathbf{i} - \frac{1}{2}\mathbf{j}$ $\overrightarrow{MC} = (3\mathbf{i} - \mathbf{j}) - \left(\frac{3}{2}\mathbf{i} - \frac{1}{2}\mathbf{j}\right)$ $= \left(\frac{3}{2}\mathbf{i} - \frac{1}{2}\mathbf{j}\right) = \overrightarrow{OM}$	B1(AO 2.1) M1(AO 1.1a) A1(AO 2.1) [3]		
		c	$\lvert\overrightarrow{MC}\rvert = \left\lvert\frac{3}{2}\mathbf{i} - \frac{1}{2}\mathbf{j}\right\rvert = \frac{1}{2}\sqrt{3^2 + (-1)^2}$ $= \frac{1}{2}\sqrt{10}$	M1(AO 1.1a) A1(AO 1.1b) [2]	Or exact equivalent, e.g. $\sqrt{\left(\frac{3}{2}\right)^2 + \left(-\frac{1}{2}\right)^2}$ Accept alternative answer $\sqrt{\frac{5}{2}}$	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
12		a	$\sec\theta - \cos\theta \equiv \dfrac{1}{\cos\theta} - \cos\theta$ $\equiv \dfrac{1-\cos^2\theta}{\cos\theta}$ $\equiv \dfrac{\sin^2\theta}{\cos\theta} \equiv \dfrac{\sin\theta}{\cos\theta}\sin\theta$ $\equiv \tan\theta \sin\theta$ AG	B1(AO 1.2) M1(AO 2.1) M1(AO 1.2) E1(AO 2.1) [4]	Use of definition of sec Manipulation of fractions Use of at least one trig identity Clear and complete argument	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	$\tan\theta\sin\theta=\frac{1}{2}\tan\theta\Rightarrow\tan\theta\left(\sin\theta-\frac{1}{2}\right)=0$	M1(AO 1.1a)	Factorising		
			$\tan\theta = 0$ gives $\theta = 0, \pi$	A1(AO 1.1b)	Both values		
			$\sin\theta=\frac{1}{2}$ gives $\theta=\frac{1}{6}\pi,\ \frac{5}{6}\pi$	M1(AO 1.1a)	At least one value from arcsin	SCI for sole answer $\frac{1}{6}\pi$ www	
				A1(AO 1.1b)	Both values correct		
			Alternative method				
			$1-\cos^2\theta=\frac{1}{2}\sin\theta\Rightarrow\sin^2\theta=\frac{1}{2}\sin\theta$	M1	Obtaining quadracit in $\sin\theta$		
			$\sin\theta = 0$, gives $\theta = 0, \pi$	A1	Both values		
			$\sin\theta=\frac{1}{2}$ gives $\theta=\frac{1}{6}\pi,\ \frac{5}{6}\pi$	M1	At least one value from arcsin		
				A1	Both values correct		
				[4]			
			Total	8			

Question			Answer/Indicative content	Marks	Part marks and guidance		
13			Gradient of chord	M1(AO 2.1)			
			$=\dfrac{(x+h)^3-x^3}{h}$ oe	M1(AO 1.1a)	Expansion of cubic attempted, with at least some terms correct	Not just $x^3 + h^3$	
			$=\dfrac{x^3+3x^2h+3xh^2+h^3-x^3}{h}$	A1(AO 1.1)			
				E1(AO 2.4)	Completely correct expansion		
			Derivative is limit of gradient of chord as $h \to 0$	B1(AO 2.2a)			
			$3x^2 + 3xh + h^2 \to 3x^2$ AG	[5]			
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance		
14		a	DR At intersections, $x^2 - kx = 3(k + 1) + kx - x^2$ $2x^2 - 2kx - 3(k + 1) = 0$ For touching, '$b^2 - 4ac = 0$' $4k^2 + 24(k + 1) = 0 \Rightarrow k^2 + 6k + 6 = 0$ $k = \frac{-6 \pm \sqrt{12}}{2}$ $k = -3 \pm \sqrt{3}$	M1(AO 3.1a) M1(AO 1.1) M1(AO 2.1) A1(AO 2.2a) B1(AO 1.1) B1(AO 1.1) [6]	Forming quadratic 0 on one side of quadratic Forming quadratic from discriminant Correct quadratic in 3-term form Use of formula or completing the square	$(k + 3)^2 = 3$	
		b	On y-axis $x = 0$, so $y = 0^2 - 0k = 0$ If they cross on the y-axis, they cross at the origin $0 = 3(k + 1) + k0 - 0^2$ $k = -1$ [so a value exists]	M1(AO 3.1a) B1(AO 2.2a) M1(AO 1.1) A1(AO 2.1) [4]	Use of $x = 0$ to find y Use of $x = 0$ and $y = 0$		
			Total	10			

Question			Answer/Indicative content	Marks	Part marks and guidance		
15			2^k evaluated for any positive integer k eg $2^4 - 1 = 15 = 5 \times 3$ which is not prime	M1(AO 1.1) A1(AO 2.4) [2]	Any positive integer which generates any non-prime		
			Total	2			

Question			Answer/Indicative content	Marks	Part marks and guidance	
16			n $n+1$□□□□□$n+2$ soi	B1	may be earned later	
			$(n+2)^2 - n^2$ soi	M1	allow ft for next three marks for other general consecutive integers eg $n-1$ □ n□□□□ $n+1$ for other integers in terms of n (eg $2n$, $2n+1$, $2n+2$ or $2n+1$, $2n+3$, $2n+5$)	allow $n^2 - (n+2)^2$ for M1 then A0 for negative answer; may still earn last B1
			$4n + 4$ obtained with at least one interim step shown	A1		
			$4(n+1)$ or $\dfrac{4n+4}{4} = n+1$	B1		
				[4]	allow ft for this M1 only	B0 for $n + 1 \times 4$
					may be obtained independently	
					Examiner's Comments The majority of candidates that attempted this standard proof question gained full marks, showing the needed interim step(s) to obtain the corresponding accuracy marks. A minority chose wrong expressions for the three integers (e.g. n, $2n$, $3n$). Unfortunately candidates missing the middle term of $4n$ when squaring the $(n+2)$ term was seen quite	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					often. Some candidates considered the first term squared minus the last term squared and then conveniently ignored the negative signs. A handful of candidates attempted an entirely numerical approach.	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance		
17			Suppose the polygon has n sides. Then $180(n - 2) = 155n$ $\Rightarrow 25n = 360$ [$\Rightarrow n = 14.4$] which is impossible as n is an integer So no regular polygon has interior angle 155° **or** When $n = 14$, int angle = 180×12/14 = 154.29° When $n = 15$, int angle = 180×13/15 = 156° So no n which gives an interior angle 155°.	M1 A1 A1cao B1 B1 B1 [3]	or sum of ext angles = 360° so $25n = 360$ or 72/5 clear statement of conclusion accept 154°		
					Examiner's Comments Candidates scored full marks or zero marks in roughly equal numbers here. Most gave the first method shown in the mark scheme, namely solving $180(n - 2)$ $= 155n$ to get n = 14.4, but we also saw some examples of the second approach, finding the interior angles for 14 and 15 sides. By far the most common error was to solve $180(n - 2) = 155$, getting n = 2.86.		
			Total	3			

Question			Answer/Indicative content	Marks	Part marks and guidance		
18			<table><tr><th>*AB*</th><th>*BC*</th><th>*AF*</th></tr><tr><td>1</td><td>2</td><td>$\sqrt{20}$</td></tr><tr><td>1</td><td>3</td><td>$\sqrt{15}$</td></tr><tr><td>2</td><td>3</td><td>$\sqrt{12}$</td></tr></table> eg none of the values of AF are integers and all possibilities have been covered.	M1(AO2.1) A1(AO1.1) E1(AO2.4) [3]	attempt at proof by exhaustion all cases covered ignore cases where AF < BC supporting comment may be brief but must refer to values of *AF*		
			Total	3			

Question			Answer/Indicative content	Marks	Part marks and guidance	
19		a	$AB^2 = 2a^2 - 2a^2 \cos\theta$	B1(AO1.1b) [1]		
		b	$AD = a\sin\left(\frac{1}{2}\theta\right)$ AB = 2AD ⇒ $AB^2 = 4AD^2$, so $2a^2 - 2a^2\cos\theta = 4\left(a\sin\left(\frac{1}{2}\theta\right)\right)^2$ $1-\cos\theta = 2\sin^2\left(\frac{1}{2}\theta\right) \Rightarrow \cos\theta = 1-2\sin^2\left(\frac{1}{2}\theta\right)$	B1(AO3.1a) M1(AO2.1) E1(AO2.1) [3]	Must be in terms of *a* and *α* Must handle squared term correctly AG Result must be clearly shown	
		c	Proved for $0 < \theta < 180°$ (as θ is angle in a triangle)	E1(AO2.3) [1]	Allow 'between' in words	Or $0 < \theta < \pi$
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
20		a	$x = -b$ $y = 0$	B1(AO2.2a) B1(AO1.1) [2]		
		b	For neg a, $\frac{a}{(x+b)^2} < 0$ ative So in this case the curve lies below the x-axis and Joe's statement is false	B1(AO2.3) E1(AO2.4) [2]		If zero scored, SC1 for $(x + b)^2$ is never negative oe
		c	$x = -b$ is a line of symmetry $b = -2$ $3 = \frac{a}{(1+b)^2}$ a = 3	M1(AO3.1a) A1(AO2.2a) M1(AO1.1a) A1(AO1.1) [4]	Or $x = 2$ is a line of symmetry O $3 = \frac{a}{(3+b)^2}$ r	Or $(1 + b)^2 = (3 + b)^3$
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
21			$(2n + 1)^2 - (2n - 1)^2$ oe	B1(AO2.1)		OR $P = Q + 2$ $P^2 - Q^2 = (Q + 2)^2 - Q^2$ B1 $= 4(Q + 1)$ (actorized) M1 $(Q + 1)$ divisible by 2 so $4(Q + 1)$ is multiple of 8 A1 OR $Q = P - 2$
			$4n^2 + 4n + 1 - (4n^2 - 4n + 1)$	M1(AO1.1)	Allow one slip eg sign error	
			$= 8n$ (so multiple of 8)	A1(AO2.4) [3]	Note: Numerical verification 0	
					<u>Examiner's Comments</u> There were quite a number of completely correct proofs, usually starting with P = $2n + 1$ or $2n + 3$. A small number of candidates put P = $2n + 1$ and Q = $2n + 3$; these candidates could gain some credit for subsequent work. A considerable number of candidates did not use the fact that P and Q were odd at the start of the proof, and worked with P and $(P - 2)$ or Q and $(Q + 2)$, or with n and $(n + 2)$ or $(n - 1)$ and $(n + 1)$. These candidates generally showed that $P^2 - Q^2$ is a multiple of 4; full credit could have been awarded by using the fact that P and Q are odd at this point to prove the required result, but this was not seen. Attempts to show this by giving numerical responses were also seen, but these	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					scored 0. Occasionally a candidate gave several numerical responses, and claimed that this was 'proof by exhaustion'.	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
22		a	9483	B1(AO1.1) [2]	BC **<u>Examiner's Comments</u>** This part was almost always done correctly.	
		b	7 × 10 112 +10259 – 10014 soi = 71029 66381 – 9204 + *x* = "71029" Emma needs to make 13852 steps	M1(AO3.1a) A1(AO1.1) M1(AO1.1) A1(AO3.2a) [4]	NB Rose's new mean is 10147 $\frac{7 \times 10\,112 + 10259 - 10014}{7}$ $= 10147$ $= \frac{66381 - 9204 + x}{7}$ Using 8 days can gain M1 only **<u>Examiner's Comments</u>** Most candidates found this part quite straightforward. One common error in this part was to use 10 112, the original 7-day mean for Rose, rather than work out the new 7-day mean for days 2 to 8 inclusive. Another common error was to find how many steps Emma should take on day 8 so that she has taken as many steps as Rose over the period of 8 days; this was often done by considering 8-day means.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	10341 + 2×948 soi = 12237 Comparison of their 13852 with their 12237 13852 is an outlier, so Emma would need to make an unusually high number of steps on day 8	M1(AO3.1b) M1(AO1.1) A1(AO3.2b) [3]	Soi Conclusion; 'outlier' not essential. Dep M2 www **Examiner's Comments** Candidates were expected to compare their answer to part (b) with the long-term mean number of steps for Emma given in the question, 10 341. In order to gain full credit candidates were expected to understand that the question was asking if the answer to part (b) was an outlier, and were expected to use the definition of outlier given in section D13 of the specification; candidates using other definitions of outlier were not given full credit. Candidates using the data in the table in the question to work out quartiles or mean and standard deviation were not given credit.	
			Total	8		

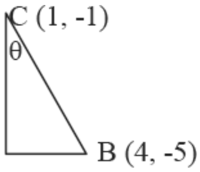
Question			Answer/Indicative content	Marks	Part marks and guidance	
23		a	C is (1, –1)	B1 (AO 1.1a) [1]	cao **Examiner's Comments** Most candidates were correct here.	

Question	Answer/Indicative content	Marks	Part marks and guidance
b	(i) EITHER Substitute	M1 (AO 3.1a)	AG Attempt to eliminate one variable
	$y=\frac{3}{4}x-8$ into the equation of the circle $(x-1)^2+\left(\frac{3}{4}x-8+1\right)^2=25$	M1 (AO 1.1a)	Attempt to expand and collect terms to obtain 3 term quadratic expression
	$x^2 = 8x + 16 = 0$	A1 (AO 1.1b)	A correct 3 term quadratic
	EITHER $(x-4)^2 = 0$ OR Discriminant = $(-8)^2 - 4 \times 1 \times 16 = 0$ So the equation has a repeated root so the line is a tangent	A1 (AO 2.2a) [4]	Clearly argued
	OR Substitute	M1 (AO 3.1a)	AG
		M1 (AO 1.1a)	Attempt to eliminate one variable
	$x=\frac{4}{3}y-\frac{32}{3}$ into the equation of the circle $\left(\frac{4}{3}y-\frac{32}{3}-1\right)^2+(y+1)^2=25$	A1 (AO 1.1b)	Attempt to expand and collect terms to obtain 3 term quadratic expression

Question			Answer/Indicative content	Marks	Part marks and guidance	
			$y^2 + 10y + 25 = 0$ EITHER $(y + 5)^2 = 0$ OR Discriminant = $10^2 - 4 \times 1 \times 25 = 0$ So the equation has a repeated root so the line is a tangent	A1 (AO 2.2a) [4] B1 (AO 1.1a) [1]	A correct 3 term quadratic Clearly argued Cao	
			(ii) $x = 4$ and $y = -5$ so B is (4, –5)		**<u>Examiner's Comments</u>** There was quite a lot of confusion in this question between the tangent to the circle at A (whose equation is not given nor required) and the line $4y = 3x - 3$ which can be proved to be the tangent to the circle at B. Some candidates set out to find the equation of the tangent at A hoping to obtain the given equation. Successful candidates solved the equation of the line and the circle simultaneously although many struggled with the fractions that resulted. **<u>Examiner's Comments</u>** There was only one mark here as the x-coordinate had normally been found in part A, so there was only the y-coordinate to find.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	$\angle CAD = \angle CBD = 90^{\circ}$ (radius is perpendicular to the tangent) Gradient of $AC = \frac{2-(-1)}{5-1} = \frac{3}{4}$ Gradient of $BC = \frac{(-1)-(-5)}{1-4} = -\frac{4}{3}$ So AC is perpendicular to BC so $\angle ACB = 90^{\circ}$ So ADBC is a rectangle Either AC = BC radius [=5] Or AD = BD equal tangents so ADBC is a square. $\angle CAD = \angle CBD = 90^{\circ}$ (radius is perpendicular to the tangent) Gradient of $AC = \frac{2-(-1)}{5-1} = \frac{3}{4}$ Gradient of BD is $\frac{3}{4}$	B1 (AO 2.1) B1 (AO 3.1a) B1 (AO 2.1) [3] B1 B1 B1 B1 B1	Allow for one or other of these angles Complete proof AG Allow for one or other of these angles Complete proof AG	Allow up to B1, B1 for any two of these three pieces of evidence. Allow the final B1 only when the proof is complete and clearly argued.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
			So AC is perpendicular to BD so ADBE is a rectangle AC = BC radius so ADBC is a square. $\angle CAD = 90^{\circ}$ (radius is perpendicular to the tangent) AC = BC radius [=5] Gradien AC $= \frac{2-(-1)}{5-1} = \frac{3}{4}$ t of Equation of $y-2=-\frac{4}{3}(x-5)$ AD is So coordinates of D are (8, –2) Hence BD = 5 and AD = 5 So ABCD is a rhombus	B1	Gradient of AD must be found from the coordintes of A and C Complete proof **<u>Examiner's Comments</u>** This proved quite challenging for many candidates. The first 2 marks were given for two facts that form part of a complete proof, and the third only when a complete proof was seen. Some candidates assumed that AD and BD were perpendicular in order to find the equation of AD and subsequently used their equation to complete their answer – but this is not a proper proof. Candidates needed to show they had used the gradient of AC or the derivative of the circle to find the gradient of the tangent at A.		

Question	Answer/Indicative content	Marks	Part marks and guidance
d	E is the point (1, -6)	B1 (AO2.1)	May be implied
	EITHER  $\theta = \arctan\left(\frac{3}{4}\right) = 0.6435$	M1 (AO 3.1a) A1 (AO3.1a)	Right-angled triangle formed and use of arctan
	OR $BE = \sqrt{(4-1)^2 + (-5-(-6))^2} = \sqrt{10}$		oe
	Cosine rule in triangle BCE $\cos BCE = \frac{5^2 + 5^2 - 10}{2 \times 5 \times 5}\left[= \frac{40}{50}\right]$ ∠BCE = 0.6435...	(M1) (A1)	Using distance BC and the cosine rule
	OR M is the midpoint of BE M is (2.5, –5.5) $BM = \sqrt{(4-2.5)^2 + (-5-(-5.5))^2} = \frac{1}{2}\sqrt{10}$		oe
	$\angle BCM = \arcsin\left(\frac{\frac{1}{2}\sqrt{10}}{5}\right) = 0.32175...$ ∠BCE = 0.6435...	(M1) (A1)	
	$\text{Area sector} = \frac{1}{2}r^2\theta = \frac{1}{2} \times 25 \times 2 \times 0.32175...$ = 8.04376	M1dep (AO1.1a) A1 (AO1.1b) [5]	Using trig in triangle BCM or ECM Allow for $\angle BCM$ Oe. Must be $\angle BCE$ Using the

Question			Answer/Indicative content	Marks	Part marks and guidance
					sector area formula FT their $\angle BCM$ **Examiner's Comments** The problem solving aspect of this question involved the need to find the angle ECB before the formula for the area of the sector could be used. Many candidates managed to write down the coordinates of E for the first mark. There were several valid methods of finding the required angle but it was often difficult to follow candidates thinking unless the correct answer was obtained. The second method mark was dependent on the first so that it was not enough just to guess an angle and use that. **AfL** Write a few words of explanation to the examiner, or clear headings, indicating your method so that you can be given a method mark even if your answer is not quite right. Two method marks and the follow-through answer mark may depend on it.
			Total	14	

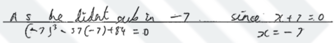
Question			Answer/Indicative content	Marks	Part marks and guidance	
24			$(a - b)^2 \geq 0 \Rightarrow a^2 - 2ab + b^2 \geq 0$ $a^2 + b^2 \geq 2ab \Rightarrow a^2 + 2ab + b^2 \geq 4ab$ $(a + b)^2 \geq 4ab$ $a+b \geq \sqrt{4ab} \Rightarrow a+b \geq 2\sqrt{ab} \Rightarrow \frac{a+b}{2} \geq \sqrt{ab}$	B1 (AO 1.1) B1 (AO 3.1a) B1 (AO 2.1) [3]	Squaring bracket Adding $2ab$ to each side Square root and correct completion **<u>Examiner's Comments</u>** Many fully correct proofs seen here. However some candidates stated that $a^2 + b^2 = (a + b)^2$.	
			Total	3		

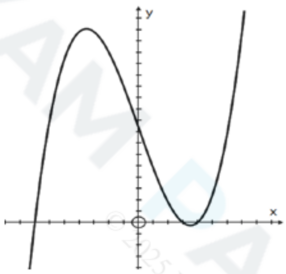
Question			Answer/Indicative content	Marks	Part marks and guidance	
25		a	Angle BDC = 90 – θ (angles of triangle) Angle CDA = θ (Angle ADB = 90° as it is the angle in a semicircle)	M1 (AO 2.1) E1 (AO 2.2a) [2]	Including reason Answer given so mark is for reason **Examiner's Comments** This question drew on prior knowledge from GCSE. Only few managed to give both 'angles in a triangle' and 'angle in a semicircle'.	Reasons can be given in either order
		b	Triangle ACD, $\tan\theta = \frac{h}{b}$ Triangle BCD, $\tan\theta = \frac{a}{h}$ $\frac{a}{h} = \frac{h}{b} \Rightarrow h^2 = ab \Rightarrow h = \sqrt{ab}$	M1 (AO 1.1) E1 (AO 2.1) [2]	At least one correct expression for $\tan\theta$ Setting expressions equal and correct completion to given answer AG **Examiner's Comments** This was generally proved correctly.	Alternative method: triangle ACD is similar to triangle DBC
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
26			Suppose that for the given perimeter, $4L$, there is a rectangle which is larger in area than the square. There is a square which has the same area as this rectangle but a smaller perimeter so its side is less than L. The square with side L has perimeter $4L$ and an area larger than the given rectangle. This is a contradiction so the square must have the largest area of all rectangles with given perimeter.	M1 (AO 2.5) A1 (AO 3.1a) A1 (AO 2.2a) [3]	Setting up a statement for contradiction. Use of statement in line 31–32 Completion to correct conclusion, including contradiction **<u>Examiner's Comments</u>** Proof by contradiction was challenging for the majority of candidates. Many offered no solution and some tried a proof by deduction. Those candidates that were successfully were able to make progress from a clear, suitable initial statement for contradiction.	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance		
27			$x^2+x+2=\left(x+\frac{1}{2}\right)^2+1\frac{3}{4}$ Minimum value is $1\frac{3}{4}$ hence always greater than 1 **Alternative solution 1** $y=x^2+x+2 \Rightarrow \frac{dy}{dx}=2x+1$ Minimum occurs when $x=-\frac{1}{2}$ Minimum value is $1\frac{3}{4}$ hence always greater than 1 **Alternative solution 2** Discriminant of $x^2 + x + 1$ is $1 - 4$ $= -3$ Discriminant negative so $x^2 + x + 1$ is never zero, when $x = 0$, $x^2 + x + 1 = 1$, hence always positive.	M1 (AO 3.1a) A1 (AO 1.1) E1 (AO 2.4) M1 A1 E1 M1 A1 E1 [3]	Attempt to complete square Correct completed square form Conclusion clearly explained Conclusion clearly explained Must be working on $x^2 + x + 1 > 0$ Complete argument	Or equivalent steps, following initial rearrangement as $x^2 + x + 1 > 0$ Or equivalent steps, following initial rearrangement as $x^2 + x + 1 > 0$	
			Total	3			

Question			Answer/Indicative content	Marks	Part marks and guidance		
28			% increase in price $= \frac{100k}{P}$ $\text{PED} = \frac{100h}{Q} \div \frac{100k}{P} = \frac{hP}{kQ}$ $\frac{k}{h} = m$ so $\text{PED} = \frac{P}{mQ}$	B1 (AO 1.1a) M1 (AO 2.2a) E1 (AO 2.1) [3]	AG Correct working needed	Lose 1 mark max for sign errors in this question	
			Total	3			

Question			Answer/Indicative content	Marks	Part marks and guidance	
29	a		DR Nigel should substitute $x = -7$ not $x = 7$; his calculation shows that $(x - 7)$ is not a factor $(-7)^3 - 37 \times (-7) + 84 = -343 + 259 + 84 = 0$, so $(x + 7)$ is a factor	B1 (AO2.3) B1 (AO2.1) [2]	Any suitable comment that references the factor theorem Clear argument needed	Allow for correct algebraic division if comment made about no remainder.
					Examiner's Comments Many candidates understood the use of the factor theorem here but not all were able to explain sufficiently clearly to obtain full marks. **Exemplar 2**  This exemplar is a case where the candidate did not have a very clear explanation of why Nigel's argument was wrong but was considered to be just enough. However, it is a very good example how the second mark can be lost if there is no conclusion given after finding the zero value.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
	b		DR $x^3 - 37x + 84 = (x + 7)(x^2 - 7x + 12)$	M1 (AO3.1a)	Attempt to divide cubic by $(x + 7)$ or find a quadratic factor by inspection	May be seen above (Allow if at least two correct terms)
			$= (x + 7)(x - 3)(x - 4)$	M1 (AO1.1a)	Attempt to factorise their quadratic factor oe and find points on x-axis FT	
			Crosses x-axis at $(-7, 0)$, $(3, 0)$, $(4, 0)$	A1 (AO1.1) (dep)	Accept values shown on sketch graph www	The A mark is dependent on the second M mark
			Crosses y-axis at $(0, 84)$	B1 (AO1.1)	Accept value shown on sketch graph www	Exact position of the stationary points is not needed
				B1 (AO1.1) [5]	Axes labelled and correct general shape FT their values as long as right way up.	
					Examiner's Comments This is a question requiring detailed reasoning, so the method marks were awarded for finding a quadratic factor of the cubic and then obtaining the linear factors leading to the points where the graph crosses the x-axis. Examiners needed to be	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					sure that the correct graph had not been obtained by simply typing the function into a graphical calculator and copying whatever they saw. The B marks for the y intercept and the general shape of the graph would have been still available but many candidates lost the final mark as they did not label their axes. (In order not to penalise the same mistake twice, this condition was not applied a second time in Q 10(a).)	
	c		**DR** Equation of the form $y = (x-1)^3 - 37(x-1) + 84$ $= (x^3 - 3x^2 + 3x - 1) - 37(x-1) + 84$ $y = x^3 - 3x^2 - 34x + 120$ **Alternative solution** Equation of the form $y = (x-1+7)(x-1-3)(x-1-4)$ $= (x+6)(x^2 - 9x + 20)$ or $(x-4)(x^2 + x - 30)$ or $(x-5)(x^2 + 2x - 24)$ $y = x^3 - 3x^2 - 34x + 120$	M1 (AO3.1a) M1 (AO1.1a) A1 (AO1.1) A1 (AO1.1) [4] M1 M1 A1 A1 [4]	Substituting $(x-1)$ for x twice Attempt to expand Correct expansion of their cubic term cao (including y =) Substituting $(x-1)$ for x everywhere Attempt to multiply out their factors Correct quadratic factor cao (including y =) <u>**Examiner's Comments**</u> Many candidates understood the change to	Allow for $(x+6)(x-4)(x-5)$

Question			Answer/Indicative content	Marks	Part marks and guidance	
					the algebra needed for a sideways translation but many candidates had their translation in the opposite direction by using f(x + 1) instead. Most had efficient ways to expand the brackets to obtain the correct expansion but the omission of “y =” cost many candidates the final mark. AfL Many candidates did not include y = in their answer despite it being included in the question and lost the final mark. In preparation for examinations, teachers need to stress the importance of accurate use of notation, and the value of checking how an answer needs to be given.	
			Total	11		

Question			Answer/Indicative content	Marks	Part marks and guidance		
30			LHS is $(\sqrt{9}\times\sqrt{2}) = \sqrt{18}$	B1 (AO2.1)	OR LHS squared is 18	No calculator. No decimal values allowed.	
			RHS is $(\sqrt{4}\times\sqrt{3}) = \sqrt{12}$	B1 (AO1.1)	RHS squared is 12		
			$\sqrt{18} > \sqrt{12}$ oe (so $3\sqrt{2} > 2\sqrt{3}$)	E1 (AO2.4)	AG	Allow proof that starts with answer, and shows it must be true.	
					OR eg $\sqrt{3}\times\sqrt{3}\times\sqrt{2} > \sqrt{2}\times\sqrt{2}\times\sqrt{3}$ $\sqrt{3} > \sqrt{2}$ which is true		
					<u>Examiner's Comments</u> The most common correct solutions were to write the left side as $\sqrt{18}$ and the right side as $\sqrt{12}$, or to square each side to get 18 and 12, and then to point out that 18 is bigger than 12. A considerable number of candidates did not know what to do, and some gave decimal values from calculators, which was not accepted. There were also some different correct solutions; an elegant one is shown in Exemplar 1.		

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance	
					Exemplar 1 $3\sqrt{2}$ $\quad$ $2\sqrt{3}$ $3\sqrt{2} \times \sqrt{6} = 3\sqrt{12} = 3 \times 2\sqrt{3} = 6\sqrt{3}$ $\underbrace{2\sqrt{3} \times \sqrt{6}} = 2\sqrt{18} = 2 \times 3 \times \sqrt{2} = 6\sqrt{2}$ $2 < 3$ $\quad$ Sub into inequality $\therefore \sqrt{2} < \sqrt{3}$ $\therefore 6\sqrt{2} < 6\sqrt{3}$ $\therefore 2\sqrt{3} \times \sqrt{6} < 3\sqrt{2} \times \sqrt{6}$ $\therefore 2\sqrt{3} < 3\sqrt{2}$	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
31			Suppose there is a largest prime, [p] Multiply all the primes up to and including [*p*] and add 1 This number is not divisible by any of the primes Therefore it is prime (and larger than *p*) This is a contradiction (so there is no largest prime)	M1 (AO2.1) A1 (AO2.3) A1 (AO2.4) [3]	Allow 'If there is …' oe Must mention 'contradiction'	
					Examiner's Comments Many candidates struggled with constructing a proof by contradiction. The first step is to assume the opposite of what you are trying to prove, then show how this leads to a contradiction and importantly to complete the proof explain the contradiction and what this then proves. Many candidates were familiar with proof started in the question and could provide the above structure to complete the proof although weaker ones tried to change it by adding 2 or subtracting 1, etc. A few of the more confident candidates knew the full proof in that the constructed number must either be prime or have a prime factor larger than the previously assumed 'largest prime'. However examiners accepted either or both of these points as adequate contradictions and were more interested in seeing	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					the 3-step structure outlined above. **Exemplar 5** ~~Let the~~ Assume there is a finite number of primes. To find the next largest prime number, multiply all the primes and add 1. The new number is not divisible by any of the primes in the list. Therefore it must be a prime number, which is a contradiction. This means that there is an infinite ~~number~~ amount of prime numbers. Possibly one of the most concise full mark answers seen. This illustrates that that fancy language and post A Level skills are not necessary.	
			Total	3		
32			When $n = 2$, $2^2 + 2 \times 2 + 10$ (= 18) which is not odd So the statement must be false (counterexample)	M1(AO2.1) A1(2.2a) [2]	Use of $n = 2$ seen Complete argument must include a clear conclusion that statement false	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance		
33	a		$a = 1$ and $b > 1 \Rightarrow a^b = 1$ and $b^a = b$ Hence $a^b < b^a$	B1 (AO1.1) E1 (AO2.2a) [2]	Convincing completion; AG	Subbing values may score B1 but not E1	
	b		Integer values of a and b with $b > a > 1$ such that ab not greater than b^a	B1 (AO2.2) [1]	Possible values • $a = 2$, $b = 3$ • $a = 2$, $b = 4$		
			Total	3			